

Chapter 4, Section 5

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1. Show that a hyperelliptic curve can never be a complete intersection in any projective space.

Proof. Combine (Ex. 3.3) and (5.2). □

2. If X is a curve of genus ≥ 2 over a field of characteristic 0, show that the group $\text{Aut } X$ of automorphisms of X is finite.

Proof. Suppose X is hyperelliptic, and let $f : X \rightarrow \mathbf{P}^1$ be the unique g_2^1 of X . If $\sigma \in \text{Aut } X$, then $f' = f \circ \sigma$ is also a g_2^1 , so by uniqueness, there exists $\tau \in \text{Aut } \mathbf{P}^1$ such that $f' = \tau \circ f$. In particular, τ sends the ramification points of f to the ramification points of f' . If $\tau = \text{id}$, then either $\sigma = \text{id}$ or σ is the automorphism interchanging the sheets of f . If $\tau \neq \text{id}$, then τ gives an element of S_{2g+2} , where $2g+2$ is the number of ramification points of f .

X is not hyperelliptic. Let $X \subseteq \mathbf{P}^{g-1}$ be the canonical embedding of X . □

3. *Moduli of Curves of Genus 4.* The hyperelliptic curves of genus 4 form an irreducible family of dimension 7. The nonhyperelliptic ones form an irreducible family of dimension 9. The subset of those having only one g_3^1 is an irreducible family of dimension 8.

Proof. Hyperelliptic elliptic curves of genus 4 are determined as two-fold coverings of $\mathbf{P}^1$, ramified at $0, 1, \infty$, and $2g-1=7$ additional points, up to the action of a certain finite group. So the hyperelliptic curves form an irreducible subvariety of dimension 7 of $\mathfrak{M}_4$.

The canonical embedding of nonhyperelliptic curves of genus 4 is a complete intersection of a unique irreducible quadric surface and an irreducible cubic surface.

Fix a quadric surface $Q \subseteq \mathbf{P}^3$. The family of all these curves is parametrized by an open set $U \subseteq \mathbf{P}(\text{H}^0(Q, \mathcal{O}_Q(3)))$, which has dimension 15. So there is a morphism $U \rightarrow \mathfrak{M}_4$ with $F_2 \in U$ mapped to the isomorphism class of the curve $F_2.Q$.

Q is nonsingular. U has dimension 15. $Q.F \cong Q'.F'$ if and only if there exists $\sigma \in \text{PGL}(3)$ such that $\sigma(Q). \sigma(F) = Q'.F'$ if and only if $\sigma(Q) = Q'$ and there exists $\tau \in \text{PGL}(3)$ such that τ fixes Q' and $\tau(\sigma(F)) = F'$. Thus, the fibre of $U \rightarrow \mathfrak{M}_4$ over $Q.F$ is the image of the subgroup $G \subseteq \text{PGL}(3)$ fixing Q , which has dimension 6. Note that $\sigma \in G$ fixing Q is equivalent to fixing a nondegenerate quadratic form. If $\sigma \in G$ fixes $Q.F$ as a subscheme of Q , then σ induces a nontrivial automorphism of $Q.F$. The automorphism group of a curve of genus ≥ 2 is finite. Hence, the fibres of $U \rightarrow \mathfrak{M}_4$ over $Q.F$ has dimension 6, so the image of $U \rightarrow \mathfrak{M}_4$ has dimension 9.

Q is singular. U has dimension 15, but subgroup $G \subseteq \text{PGL}(3)$ fixing Q has dimension 7. □

5. *Curves of Genus 5.* Assume X is not hyperelliptic.

- (a) The curves of genus 5 whose canonical model in $\mathbf{P}^4$ is a complete intersection $F_2.F_2.F_2$ form a family of dimension 12.
- (b) X has a g_3^1 if and only if it can be represented as a plane quintic with one node. These form an irreducible family of dimension 11.
- (c) In that case, the conics through the node cut out the canonical system (not counting the fixed points at the node). Mapping $\mathbf{P}^2 \rightarrow \mathbf{P}^4$ by this linear system of conics, show that the canonical curve X is contained in a cubic surface $V \subseteq \mathbf{P}^4$, with V isomorphic to $\mathbf{P}^2$ with one point blown up (II, Ex. 7.7). Furthermore, V is the union of all the trisecants of X corresponding to the g_3^1 (5.5.3), so V is contained in the intersection of all the quadric hypersurfaces containing X . Thus V and g_3^1 are unique.

Proof.

- (a) $\dim \text{Stab}(Q) = 15$, $\dim \mathbf{P} H^0(Q, \mathcal{O}_Q(2)) =$
- (b) Suppose there exists a birational morphism $f : X \rightarrow X'$ such that X' is a plane quintic with one node. Then f is the normalization of X' , and the preimage of the node is a divisor D of degree 2 on X .
- (c)

□

- 6. Show that a nonsingular plane curve of degree 5 has no g_3^1 . Show that there are nonhyperelliptic curves of genus 6 which cannot be represented as a nonsingular plane quintic curve.

Proof.

□

- 7. (a) Any automorphism of a curve of genus 3 is induced by an automorphism of $\mathbf{P}^2$ via the canonical embedding.
- (b) Assume $\text{char } k \neq 3$. If X is the curve given by

$$x^3y + y^3z + z^3x = 0,$$

the group $\text{Aut } X$ is the simple group of order 168, whose order is the maximum $84(g - 1)$ allowed by (Ex. 2.5).

- (c) Most curves of genus 3 have no automorphisms except the identity.

Proof.

- (a)
- (b)
- (c)

□